

InStore Optima - Development Session Summary

Overview

This document summarizes all major features and improvements implemented in the InStore Optima inventory management system during this development session.

Features Implemented

1. Stock Movement Tracking & Auditing

Problem: Stock edits were not being recorded in the movement audit trail.

Solution:

- Implemented automatic movement recording when stock quantities are edited
- Distinguishes between two types of movements:
 - **WRITE_OFF**: When stock is decreased (damage, expiry, loss)
 - **ADJUSTMENT**: When stock is increased (count correction)
- Added `RecordOnlyAsync()` method in `StockMovementRepository` to prevent double-updating stock values
- Backend tracks the exact difference between old and new stock values

Files Modified:

- `StockController.cs` - Updated `PUT /api/stock/{id}` endpoint
- `StockMovementRepository.cs` - Added `RecordOnlyAsync()` method
- `IStockMovementRepository.cs` - Added interface declaration

2. Bug Fix: Stock Double-Update Issue

Problem: When editing stock, the value would sometimes become incorrect (e.g., editing to 150 would save as 1).

Root Cause: The `CreateAsync()` method in `StockMovementRepository` was treating ADJUSTMENT movements as absolute values instead of relative changes, causing double-updates.

Solution:

- Separated concerns: `CreateAsync()` for auto-adjusting stock, `RecordOnlyAsync()` for logging only
- `StockController` now uses `RecordOnlyAsync()` to avoid duplicate modifications
- Movement quantities now correctly represent the difference amount

3. Replenishment Workflow Enhancement

Problem: PO (Purchase Order) popup appeared when creating replenishment, not when approving it.

Solution:

- Refactored `Replenishment.jsx` to trigger PO creation popup only after approval
- Added comprehensive PO form with:
 - **Supplier Selection:** Dropdown populated from suppliers list
 - **Expected Delivery Date:** Date picker (minimum = today)
 - **Form Validation:** Prevents submission without supplier
 - **Success Notification:** Toast confirmation after PO creation

Files Modified:

- `Replenishment.jsx` - Added `showPoPrompt`, `pendingReplId`, `poForm` state management

4. User Profile Dropdown Panel

Problem: No way to see user details or sign out from the dashboard.

Solution:

- Created profile dropdown in navbar showing:
 - User avatar with name/role header

- Email address
- User ID
- User role
- Sign-out button (red, prominent)
- Click outside to close
- Integrated with existing authentication system

Styling:

- Dark navy background (#0d1b2e)
- Light text with proper contrast
- Dark theme text colors: name #e2e8f0 , details #cbd5e1
- Light theme support with white background

5. Dark Mode Text Visibility Fix

Problem: Placeholder text in search bars and form inputs was nearly invisible in dark mode.

Solution:

- Changed placeholder color from #475569 (too dark) to #94a3b8 (readable medium grey)
- Maintains accessibility standards while being visually appropriate
- Applied to all `.form-control-custom` elements

Impact: Forms and search functionality now much more usable in dark theme

6. Comprehensive Zoom System (Accessibility)

Problem: Users with vision impairments had no way to enlarge the interface.

Solution:

- Implemented app-level zoom from 70% to 150% in 10% increments
- **Key Features:**
 - Zoom controls in user profile dropdown (-/100%/+ buttons)
 - Preset zoom options (70%, 85%, 100%, 115%, 130%, 150%)
 - Zoom applied to content only (navbar stays at 100% for usability)
 - Smooth scaling using CSS `transform: scale()`

- Persistent storage in localStorage

Browser Zoom Handling:

- Detects when user changes browser zoom using `window.devicePixelRatio`
- Auto-resets app zoom to 100% to prevent stacking
- Shows toast notification when browser zoom is detected
- Prevents confusing combined zoom levels

Files Modified:

- App.jsx - Zoom state management and browser detection
- TopNavbar.jsx - Profile dropdown with zoom controls
- TopNavLayout.jsx & InnerLayout.jsx - Prop threading

7. Public Page Zoom Controls (Landing, Login, Register)

Problem: Zoom controls were only available on the dashboard after login.

Solution:

- Created reusable `ZoomControl.jsx` component
- Added zoom controls to:
 - **Landing Page** - In top navigation bar
 - **Login Page** - Top-right corner with theme toggle
 - **Register Page** - Top-right corner with theme toggle
- Maintains consistent look and feel across all pages
- Dropdown with same presets and functionality

Component Features:

- Search/magnifying glass icon (✓ plus symbol) for zoom
- Responsive dropdown positioning
- Dark and light theme support
- Keyboard accessible

Files Created:

- `ZoomControl.jsx` - Standalone component for public pages



Technical Details

Backend Architecture Changes

```
StockController (PUT /api/stock/{id})
├─ Read existing stock
├─ Update stock via _repo.UpdateAsync()
├─ Record movement via _movementRepo.RecordOnlyAsync()
└─ Return updated stock + movement
```

Frontend State Management

```
App.jsx (Root)
├─ zoom state (70-150%)
├─ browserZoomDetected state
├─ Browser zoom detection listener
└─ Content wrapper transform: scale()
    └─ TopNavLayout
        └─ TopNavbar (zoom controls)
    └─ InnerLayout
        └─ Public Pages (Landing, Login, Register)
```

Data Flow

```
User edits stock → StockController captures difference
→ RecordOnlyAsync() logs movement → Stock + movement saved
→ No double updates → Accurate audit trail
```



UI/UX Improvements

1. Profile Dropdown

- Professional design with navy background
- Clear information hierarchy
- Easy access to sign-out

2. Zoom Controls

- Icon clearly represents zoom function
- Intuitive -/+/reset buttons
- Quick-select presets
- Smooth visual feedback

3. Dark Mode

- Improved text readability
- Consistent color scheme
- All new components support both themes

4. Form Inputs

- Placeholder text now visible
- Better visual feedback
- Improved accessibility

Testing & Validation

- **Build Status:** ✓ All builds successful (Vite: 634ms, .NET: 0 errors)
- **Stock Movements:** ✓ Write-off and adjustment correctly recorded
- **PO Workflow:** ✓ Popup appears only after approval
- **Profile Dropdown:** ✓ Opens/closes correctly, all data displays
- **Zoom System:** ✓ Works on all pages, localStorage persists
- **Browser Zoom:** ✓ Detected and handled properly
- **Dark Mode:** ✓ Text visible in all components

Files Modified/Created

Created Files

- `instore_optima_client/src/components/ZoomControl.jsx` - Zoom control component
- `instore_optima_client/src/context/MessagesContext.jsx` - Internal messaging
- `instore_optima_client/src/pages/Messages.jsx` - Messages page
- `instore_optima_client/src/services/internalMessageService.js` - Message API

Modified Frontend Files

- App.jsx - Zoom state management
- TopNavbar.jsx - Profile dropdown + zoom controls
- TopNavLayout.jsx - Prop threading
- InnerLayout.jsx - Prop threading
- Login.jsx - Zoom controls added
- Register.jsx - Zoom controls added
- LandingPage.jsx - Zoom controls added
- Replenishment.jsx - PO workflow refactor
- index.css - All styling for new components

Modified Backend Files

- StockController.cs - Stock movement recording
- StockMovementRepository.cs - RecordOnlyAsync() method
- IStockMovementRepository.cs - Interface update



Deployment Considerations

1. **Database Migrations:** Internal message system migrations included
2. **localStorage:** Users' zoom preferences will persist
3. **Browser Compatibility:** Works with all modern browsers supporting CSS transforms
4. **Accessibility:** WCAG compliant zoom controls and dark mode



Commit Information

Commit Hash: 94c9048

Branch: dev1

Date: May 30, 2026

Files Changed: 51

Insertions: 5672

Commit Message

feat: comprehensive inventory & accessibility improvements

- Stock movements: Record WRITE_OFF (decrease) and ADJUSTMENT (increase) movements
- Fix stock double-update bug: Use RecordOnlyAsync()
- Replenishment workflow: Auto-generate PO popup after approval
- Profile dropdown: Display user details with sign-out option
- Dark mode text fixes: Improved placeholder visibility
- Zoom system: Accessible zoom controls (70-150%) with presets
- Browser zoom detection: Auto-reset app zoom with notification
- Public page zoom: Add zoom controls to Landing, Login, Register
- CSS enhancements: Dark/light theme support for all components

Key Improvements Summary

Feature	Before	After
Stock Auditing	No movement recorded on edit	Complete audit trail with WRITE_OFF/ADJUSTMENT
Stock Values	Could become incorrect	Accurate with RecordOnlyAsync()
PO Creation	Appeared at wrong time	Triggered only after approval
User Info	No profile access	Full profile dropdown
Dark Mode	Unreadable placeholders	Clear, visible text
Vision Support	No zoom capability	70-150% with presets
Public Pages	No zoom	Zoom on all pages

Related Documentation

- PROJECT_DOCUMENTATION.md - Full system documentation
- EXCEPTION_HANDLING.md - Exception handling patterns

- Database schema in `database/schema_v2.sql`

Session Complete: All requested features implemented, tested, and deployed to dev1 branch.